Who Counts as a Neighbor?

From geographic maps to a Python laboratory of proximity graphs

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How should we connect the dots? Nearest neighbors seem obvious until we ask what *near* means. Two hospitals can be geographically close but separated by a river; two measurements can be near in Euclidean distance but lie on different folds of a manifold. A point cloud contains locations, not edges. The choice of which edges to draw is a mathematical model—and it can change the conclusions.

From maps to rules for neighborhood

Geometric partitions long preceded today's graph algorithms. Voronoi's work in 1908 formalized regions closest to a given site; Delaunay triangulation later supplied a complementary account of which sites share a boundary. An evocative historical setting is John Snow's 1854 map of cholera deaths around London water pumps. *Snow did not compute a Voronoi diagram*: the modern overlay in Figure 1 asks what nearest-pump territories would look like on digitized historical coordinates. It is a geometric illustration, not a reconstruction of access routes or an epidemiological inference.

Figure 1 · John Snow's pumps: Voronoi and Delaunay

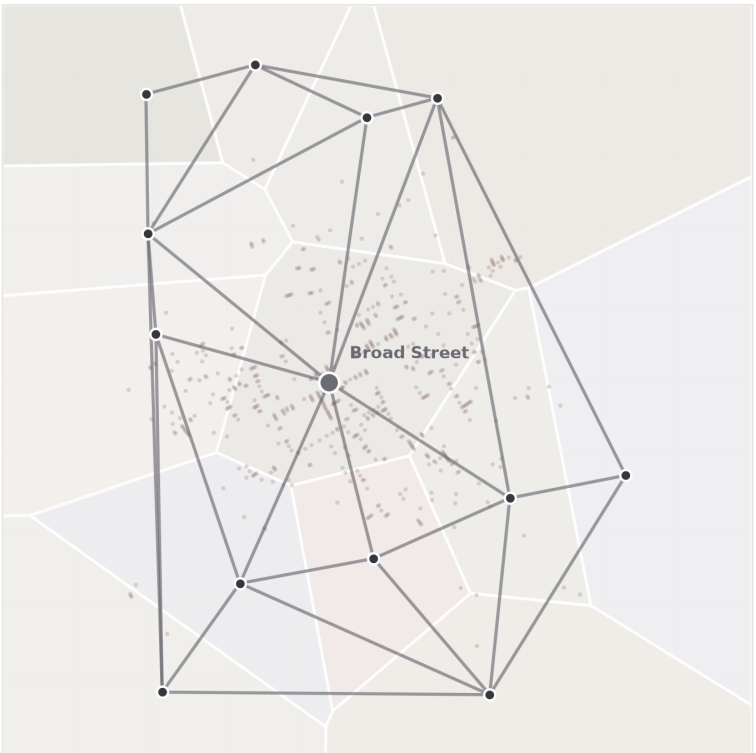


Figure 1: Historical coordinates, modern geometry. Thirteen digitized Soho pump locations define Euclidean Voronoi cells and Delaunay adjacencies; 578 recorded cholera death marks are contextual, not graph vertices. Sites: HistData digitization. Illustration: the authors using ProximityGraphs. Explore the interactive figure.

In 1969, **Gabriel and Sokal** sought a principled way to join geographically sampled biological populations. Their answer was strikingly simple: connect two sites only when the disk having their segment as diameter contains no other sample point. **Toussaint (1980)** replaced that disk with a lens-shaped region, giving the *relative neighborhood graph* (RNG). These were not just drawing conventions: an edge encoded a falsifiable geometric statement about what lay between its endpoints.

The family grew rapidly. **Edelsbrunner, Kirkpatrick, and Seidel (1983)** introduced alpha-shapes to reveal structure obscured by a convex hull. **Kirkpatrick and Radke (1985)** organized parameterized notions of point-set morphology into the β-skeleton family. **Veltkamp (1992)** went further: the two-parameter γ-neighborhood construction placed several established graphs in a wider continuum. Different rules now expressed different understandings of locality rather than competing claims to a single *correct* graph.

Why the choice of an edge matters

For generic planar samples, the classical inclusion chain is

$$\mathrm{MST} \subseteq \mathrm{RNG} \subseteq \mathrm{GG} \subseteq \mathrm{Delaunay}.$$

Moving right generally makes a graph denser; moving left preserves fewer candidate connections. In spatial infrastructure, that changes construction cost and travel detours (Aldous and Shun, 2010). In computer vision or microscopy, it changes which gaps and boundaries survive. Indeed, Maravillo Gómez and colleagues (2023) used β-skeletons to recover cyclic tiling patterns and study street-like structures.

Figure 2 makes scale visible. A convex hull encloses the entire letterform but fills its holes. An alpha-shape retains Delaunay triangles below a selected circumradius, producing straight segments; the related alpha-hull replaces boundary segments with circular arcs. As the scale moves, disconnected pieces and enclosed counters can appear or disappear. The geometry matters in shape reconstruction and motivates, but does not replace, topological data analysis.

Figure 2 · Image-traced serif Pg · alpha-shape counters

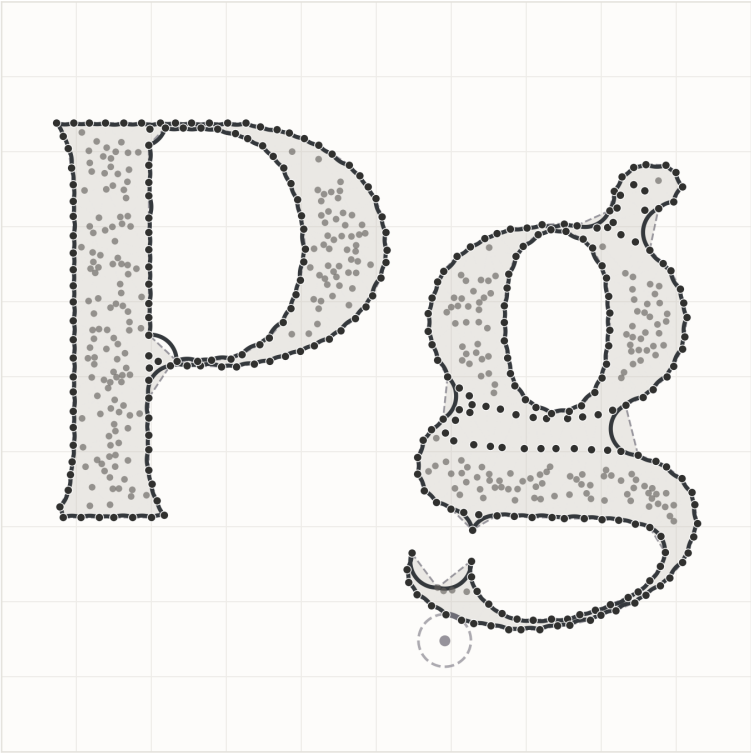


Figure 2: Shape from points. A synthetic 600-site *Pg* point cloud demonstrates the straight alpha-shape boundary, circular alpha-hull arcs, and two enclosed counters at $\alpha \simeq -28.57$. An interactive smiley and point-density controls offer other experiments. Illustration and sampling: the authors. Explore the interactive figure.

In data science, each observation may instead be a vector of gene-expression measurements or a learned image embedding. Gabriel neighborhoods have even served as an initial graph for adaptive manifold learning (Dyballa and Zucker, 2023). The general mathematics extends to $\mathbb{R}^d$, although the current ProximityGraphs software is strongest in the plane; not every constructor is dimension-general. Nor does a geometric neighborhood alone implement a graph neural network or a nearest-neighbor search service.

A probabilistic surprise

Random points do not make the geometry arbitrary. Devroye (1988) investigated expected sizes of empty-region graphs. For a stationary Poisson process of intensity ρ, suppose the exclusion region of two points at separation r has volume $a_K r^d$; then its emptiness probability is $e^{-\rho a_K r^d}$. Writing v_d for the volume of the unit d-ball yields

$$\Pr(\text{edge} \mid r) = e^{-\rho a_K r^d}, \qquad \mathbb{E}[\deg] = \frac{v_d}{a_K}.$$

Under these assumptions, the expected degree is *independent of point intensity*: it is the geometry of the forbidden region that fixes the constant. In the plane, this gives mean degree 4 for Gabriel graphs and about 2.56 for RNGs. Such predictions make proximity networks experimentally testable, rather than merely visually suggestive.

Figure 3 · Relay corridor: beta-skeleton links

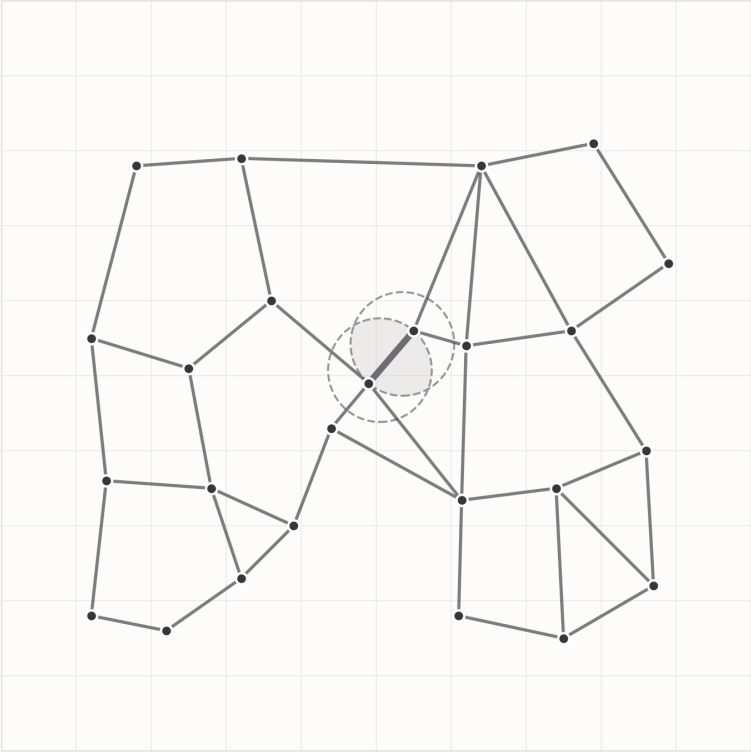


Figure 3: Edges justified by emptiness. Twenty-five synthetic sensor sites form two clusters joined by relays. The β-skeleton inspector reveals each pair's defining disks, their overlapping lune, and points that block an otherwise plausible edge. Illustration: the authors using ProximityGraphs. Explore the interactive figure.

From mathematical families to executable experiments

The open-source **ProximityGraphs Python library** brings these traditions together. Its public API spans Voronoi and Delaunay constructions; minimum-spanning, nearest-neighbor, Gabriel and relative-neighborhood graphs; β-skeletons, γ-neighborhoods, stepping-stone and elliptic-Gabriel variants; and alpha boundaries. Rather than advertising a universal best graph, the software encourages *controlled comparisons*: use the same points, change the neighborhood rule, and inspect connectivity, degree, length, shape, and robustness.

The distinction also reaches beyond empty regions. A separate *bio-inspired* module includes `PhysarumGraph`, which reinforces simulated flow-carrying links, and `FungalGraph`, which grows and prunes a sparse network heuristically. Inspired by adaptive transport in living systems (Tero and colleagues, 2010), these are algorithms to investigate, not claims that an organism has been reproduced or an optimal network found.

With `SetPoints` and `Experiment`, researchers can generate structured or stochastic samples, vary graph families and parameters, calculate metrics, and repeat simulations under documented seeds. The MIT-licensed package is available

through PyPI (`pip install proximitygraphs`); examples, source, and the documentation are open. Our interactive companion offers a no-installation entry point: drag sites, reveal the circles behind an alpha-hull or a rejected β-edge, and compare graph families on identical coordinates.

What began as a question about geographic neighbors is now a family of computational decisions spanning geometry, stochastic processes, and network science. **ProximityGraphs makes those decisions explicit, reproducible, and open to experimentation.**

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